An Analysis of the Role of Artificial Intelligence in the Personalized Development of Instrumental Music Learning in Colleges and Universities

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Abstract

Due to significant individual differences among students, instrumental music learning in colleges and universities requires precise teaching support. With the aid of artificial intelligence technology, tools such as intelligent evaluation, content recommendation, and virtual performance can effectively enhance learning efficiency and musical expressiveness. Based on practical cases in higher education, this study explores the specific application effects of AI in tracking students' learning paths and adapting teaching content. Results show that technical assistance significantly improves pitch accuracy, practice motivation, and the quality of personalized feedback, demonstrating broad prospects for promotion and in-depth application.